Software Engineering & Project

Snapshot Week 9 - BLOCK UG-16

Maptek, Block Model Compression Algorithm

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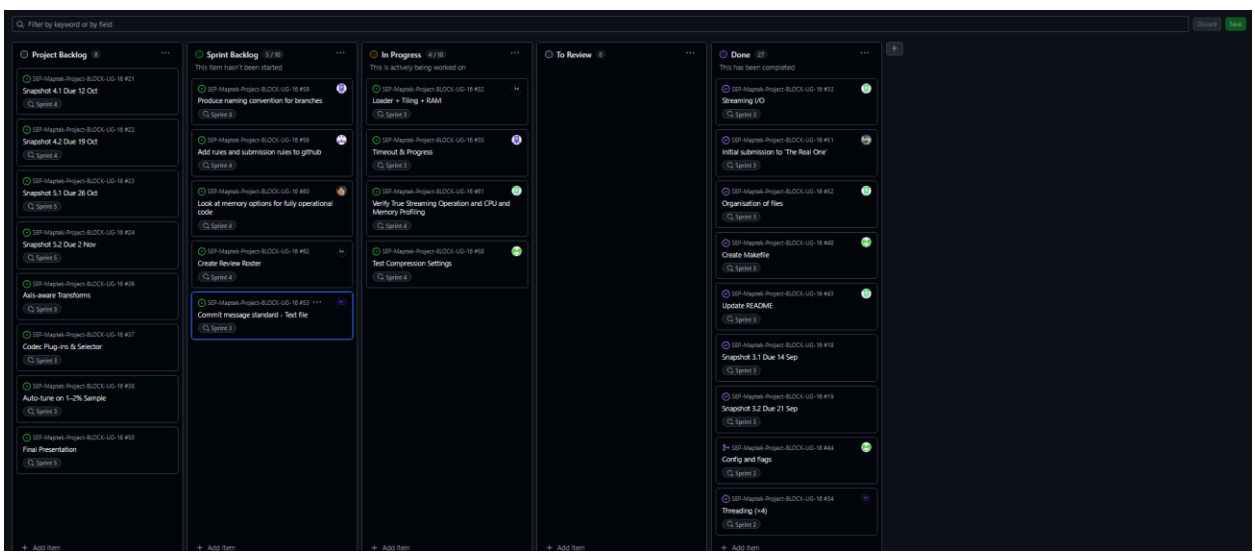
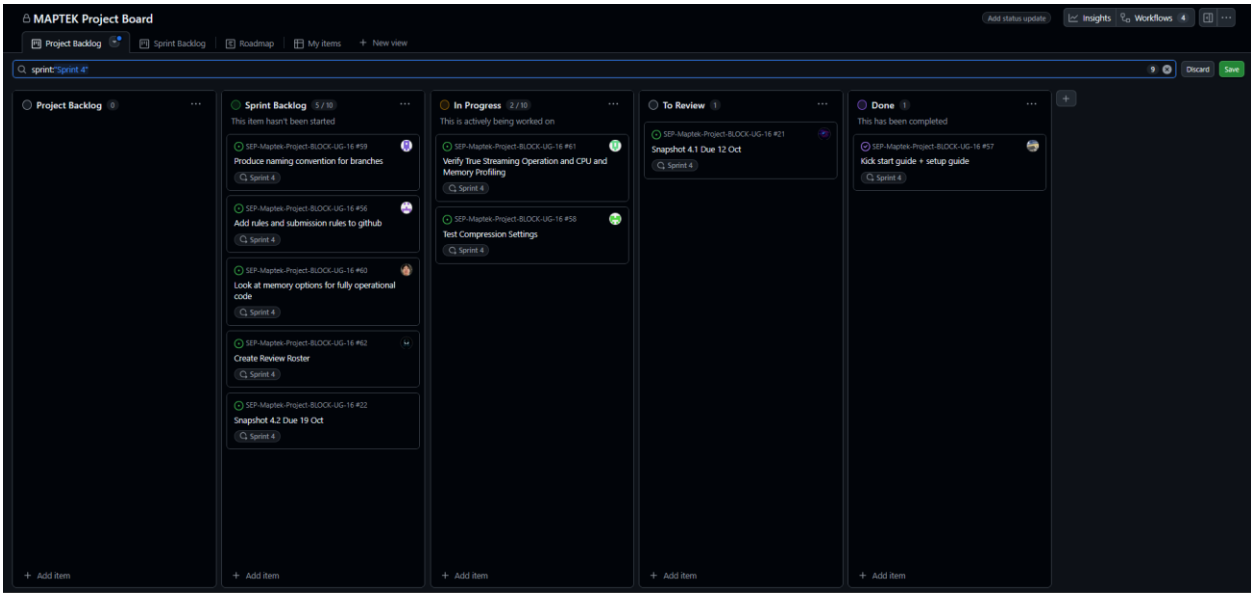
Product Backlog and Task Board

Product Backlog	
Task	**Description**
Snapshot 5.1	Create snapshot 5.1
Snapshot 5.2	Create snapshot 5.2
Axis-aware Transforms	delta/RLE per axis Palette/dictionary per tile
Correctness and Regression	Do: run current code on intro and fast competitions; confirm outputs and acceptance still pass. Done when: both older comps pass their acceptance checks.
Docs & Results Sheet	Shared results sheet (commit, params, ratio, MB/s, RAM, runtime)
Final Presentation	Initial planning for final presentation



Sprint Backlog and User Stories

Sprint Backlog	
Task	**Description**
Verify True Streaming Operation and CPU Profiling	Verify compressor reads block by block and find where time and resources are being spent throughout code
Create Kick Start and Setup Guide	Make a kick start guide in the read me and setup guide
Check Memory Options	Find variables in main code and change to reduced option
Create Review Roster	Create review roster for branches being merged to main
Produce naming convention for branches	Create naming convention file for branches
Test Compression Settings	Reduce compression settings and create either a verbal report or written report
Snapshot 4.1	Create snapshot 4.2
Snapshot 4.2	Create snapshot 4.2
Research other Githubs	Research and show other githubs with decent structure



User Stories (Sprint 4)

For a user of the block model compression algorithm, the algorithm must now handle two new challenges — *The Streaming One* and *The Worldly One* — each testing different aspects of the system.

In this sprint, the team focused on ensuring the algorithm can operate as a true stream processor capable of reading and processing data block-by-block without preloading entire datasets. This improvement enables real-time handling of infinite block models as required by *The Streaming One*. Profiling tools were used to identify CPU and memory bottlenecks, allowing targeted optimisations that improve throughput and efficiency.

In preparation for *The Worldly One*, where compression ratio is prioritised over speed, the team began testing new axis-aware transforms (delta/RLE per axis) and dictionary-based palette methods to push compression limits on large, high-resolution block models.

To support team collaboration and ensure long-term maintainability, documentation and workflow updates were also prioritised. A Kick Start and Setup Guide was created to help streamline onboarding, and a naming convention and review roster were introduced to formalise branch management and code review processes.

As the Product Owner (TA) emphasised, the sprint goal was to build both performance stability and scalability — making the algorithm robust enough for streaming environments while positioning it for maximum compression in future competitions.

Definition of Done

- The algorithm successfully processes data streams block-by-block without preloading entire datasets.
- Verified compression correctness against prior competition datasets (Intro and Fast).
- Initial axis-aware transform and palette/dictionary compression methods implemented.
- CPU and memory profiling reports produced and reviewed.
- Kick Start Guide and branch naming conventions added to documentation.

Summary of Changes

During this sprint, key progress was achieved in improving streaming reliability and compression flexibility:
Verified true streaming operation to meet The Streaming One requirements.
Conducted CPU profiling and memory analysis to locate and address major bottlenecks.
Implemented preliminary axis-aware and dictionary compression transforms to prepare for The Worldly One.
Completed new Kick Start & Setup Guide to help streamline testing and deployment.
Added review roster and branch naming conventions to strengthen team collaboration and maintain code integrity.
By the end of Sprint 4, the team has achieved stable streaming performance, clearer workflows, and foundational enhancements for the upcoming high-compression challenge. The focus now shifts toward maximising compression ratios for *The Worldly One* while maintaining efficiency and correctness.